Gap-confined diamond nanobeam optomechanical cavity with a cross phonon shield

SnV (619 nm) optical mode and 9.8 GHz breathing mode; diamond t = 120 nm; 60 nm minimum feature. COMSOL unless marked "Dmitry cloud" (laghai/nanophotonics-cloud). Ideal geometry, vertical sidewalls, isotropic diamond (E 1050 GPa, ν 0.07, ρ 3500 kg/m³, n 2.41).

1. Best design

quantity	value
beam width / thickness	650 nm / 120 nm
mirror cell	a 189.5 nm, hole hx 75.8 nm (hx/a 0.40), hy 390 nm (hy/w 0.60)
taper (Chan 2012 thesis form)	M 10 cells, d 0.086, η 1, ξ 1.79, χ -6.97; 10 mirror cells per side
optical mode	619.31 nm, Q 3.8e+06
breathing mode	9.813 GHz (localization 0.99, sidewall ratio 1.00)
mirror breathing-class (Y+Z+) gap	7.47–10.87 GHz; mode sits 1.05 GHz below the top
second localized mode	10.57 GHz (also in the gap)
m_eff / x_zpf	172 fg / 2.23 fm
zero-point strain (peak, midplane)	volumetric 1.5e-08, εxx – εyy 1.8e-08
smallest features	bridge 70, rail 97, hole 76 nm

Other gap-confined designs from the same study (all 60 nm, all with the breathing mode inside the mirror gap):

design	w	a	d, ξ, χ	λ (nm)	optical Q	breathing (GHz)	gap (GHz)	margin
w650_m186_c455_d086	650	189.5	0.086, 1.79, -6.97	619.31	3.8e+06	9.81	7.47–10.87	1.05
w650_m186_c455_d118	650	195.4	0.118, 1, -4.59	619.37	2.8e+06	9.83	7.46–10.87	1.04
w650_m192_hy66_c455_d146	650	201.2	0.146, 1.2, -2.46	619.34	2.4e+06	9.74	6.52–10.77	1.02
w700_m186_c490_d118	700	194.2	0.118, 1, -4.59	619.48	2.3e+06	9.08	6.95–10.09	1.02
w700_m198_c420_d172	700	202.3	0.172, 1.15, -1.41	619.28	8.9e+05	9.24	6.94–10.09	0.85
w700_m192_c420_d146	700	197.1	0.146, 1.57, -1.93	619.38	8.7e+05	9.23	6.95–10.09	0.86
w700_m186_c420_d118	700	191.8	0.118, 2.22, -2.73	619.52	5.9e+05	9.23	6.95–10.10	0.87

2. How the design was made

Requirement. The breathing mode must sit inside the mirror cell's breathing-class (Y+Z+) band gap, so the mirror confines it the way it confines light. The earlier best devices put the breathing mode above that gap.

Cavity form. A 1D nanobeam with elliptical holes. Cells j = 0 (center) … M taper from the defect to the mirror cell, then n_mirror plain mirror cells (Chan PhD thesis 2012, eqs. 3.48–3.53, the form Oh 2026 used): $a_j/a_m = 1 - d(2y^3 - 3y^2 + 1)$, $y = j/M$; hole area $1 - s_j = \chi(1 - a_j/a_m)$; aspect $r_j = (a_j/a_m)^{\xi}$. $\chi < 0$ makes the center holes LARGER than the mirror holes.

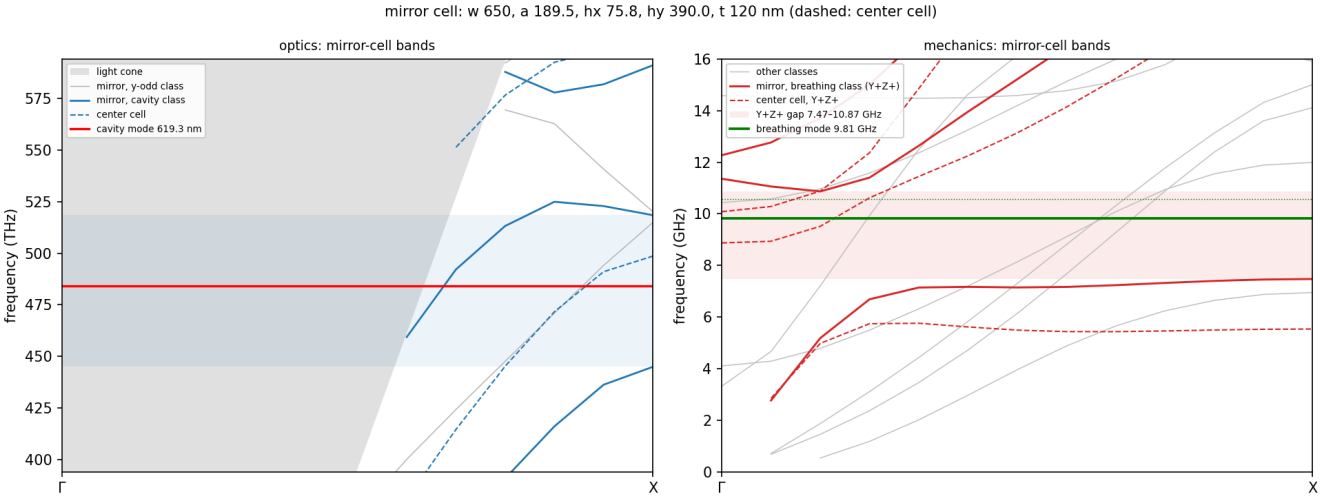
Unit-cell rule (gap-cell study, 336 cells). (1) A small-hole mirror (hx/a 0.40, hy/w 0.60) raises the top of the Y+Z+ gap to ~10.9 GHz at w 650. (2) Large center holes ($\chi < 0$) soften the center cell and pull its Γ breathing band down. The cavity breathing mode lands at the center-cell Γ frequency minus ~0.23 GHz (calibrated on 72 full cavities, rms 0.24 GHz), so pairs were kept only when both near-degenerate center-cell Γ modes, minus 0.23 GHz, sit inside the mirror gap with margin. The pair gives (d, ξ, χ) directly: $1 - d = a_c/a_m$, $s = hx_c \cdot hy_c/(hx_m \cdot hy_m)$, $\chi = (1 - s)/d$, $\xi = \ln r / \ln(1 - d)$.

Full-cavity solve. 1/8 COMSOL model (mirror planes x, y, z = 0), optics in an air box with PML (TE-like class, Ey odd about the center hole), mechanics with the beam end clamped. a_m was stepped until the cavity resonance bracketed 619.4 nm, then one solve at the interpolated a. Breathing mode = localized (≥ 0.8 of $|u|^2$ within 1.5 μm), in-phase sidewalls ($B > 0.5$), sidewall/axial amplitude ratio $S \geq 0.7$; the deformed shape was inspected.

Mirror gaps were recomputed with 13 k-points: the gap top comes from a band-2 dip near $k \approx 0.25\ \pi/a$, not Γ.

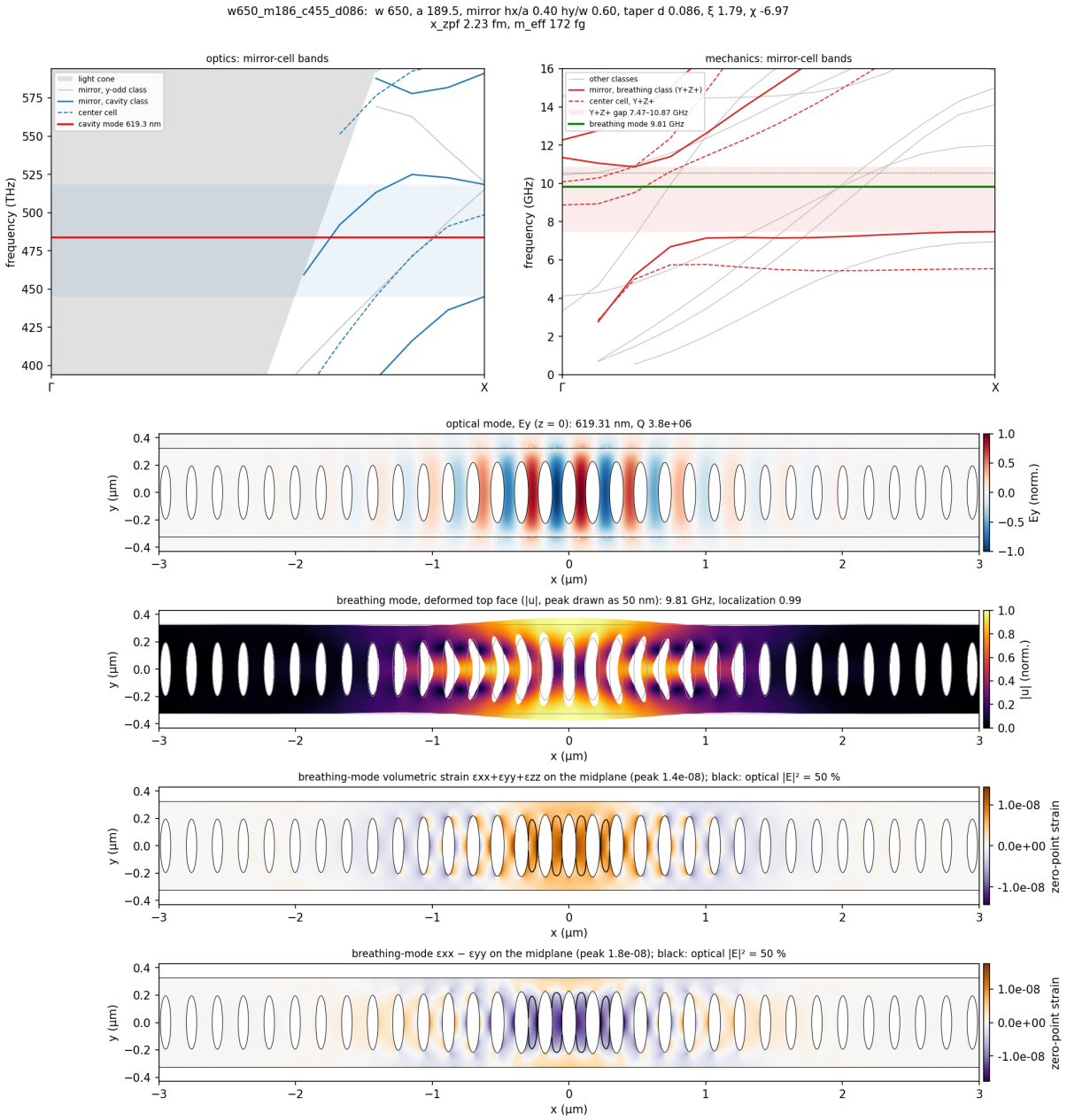
Selection. All seven designs passed; the table is sorted by optical Q. The best has the highest Q (3.8e6), a 1.05 GHz margin and 70 nm narrowest bridge. Price paid for gap confinement: optical Q 3.8e6 vs 1–2e7 for the earlier non-gap-confined devices.

3. Band diagrams



Mirror-cell bands (solid) and center-cell bands (dashed). Left: optics, Γ→X, guided bands below the light line; red line = cavity mode at 619.3 nm inside the cavity-class gap. Right: mechanics, all four symmetry classes; red = breathing class (Y+Z+); shaded = Y+Z+ gap 7.47–10.87 GHz; green = cavity breathing mode 9.81 GHz.

4. Full cavity: optical mode, breathing-mode displacement and strain



Full cavity (COMSOL). Top: band diagrams as in section 3. Then: optical Ey at z = 0; breathing mode on the deformed top face (color |u|); breathing-mode zero-point volumetric strain εxx+εyy+εzz and shear-type εxx–εyy on the midplane (scaled to x_zpf = 2.23 fm), with the optical |E|² = 50 % contour in black. Strain peaks inside the optical mode, in the bridges next to the center hole.

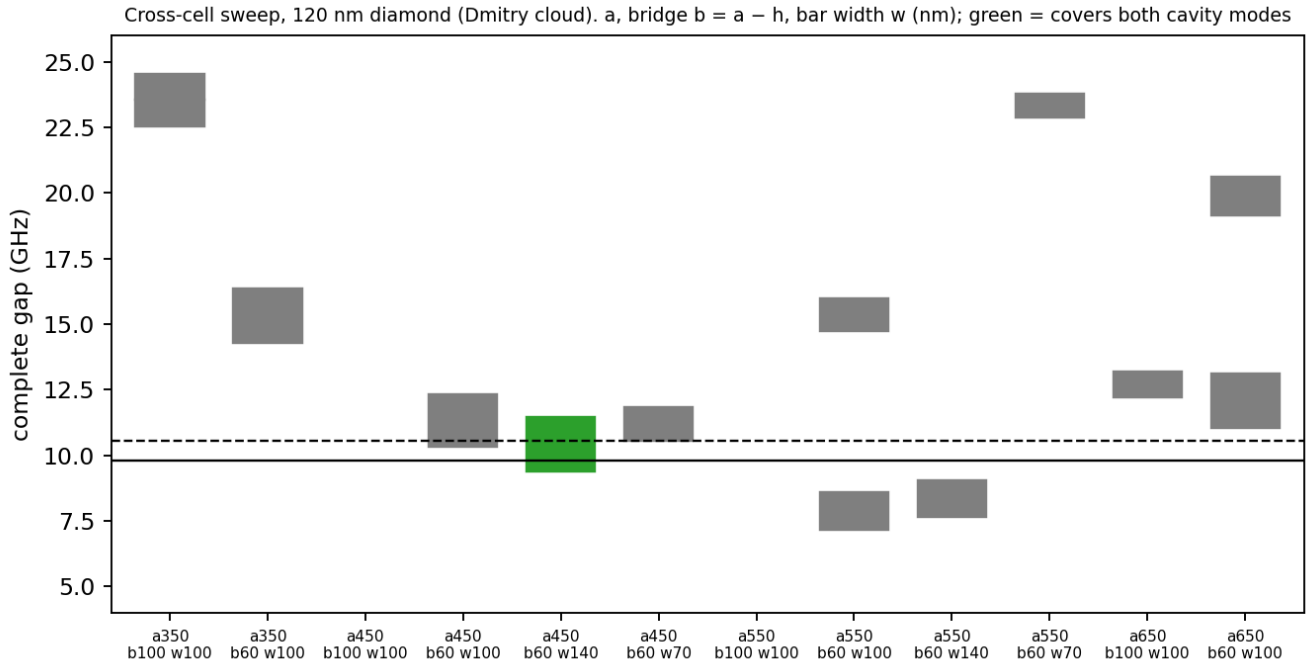
5. Square-lattice cross phonon shield

Why. The mirror confines the breathing mode only within its own symmetry class (Y+Z+) and only along the beam. Where the beam lands on the chip, the slab carries waves of every polarization in every direction. A 2D phononic crystal with a COMPLETE gap around the anchor (Chan et al., APL 2012; Oh et al. 2026 use square-lattice crosses) blocks all of them, including waves that disorder converts into other classes.

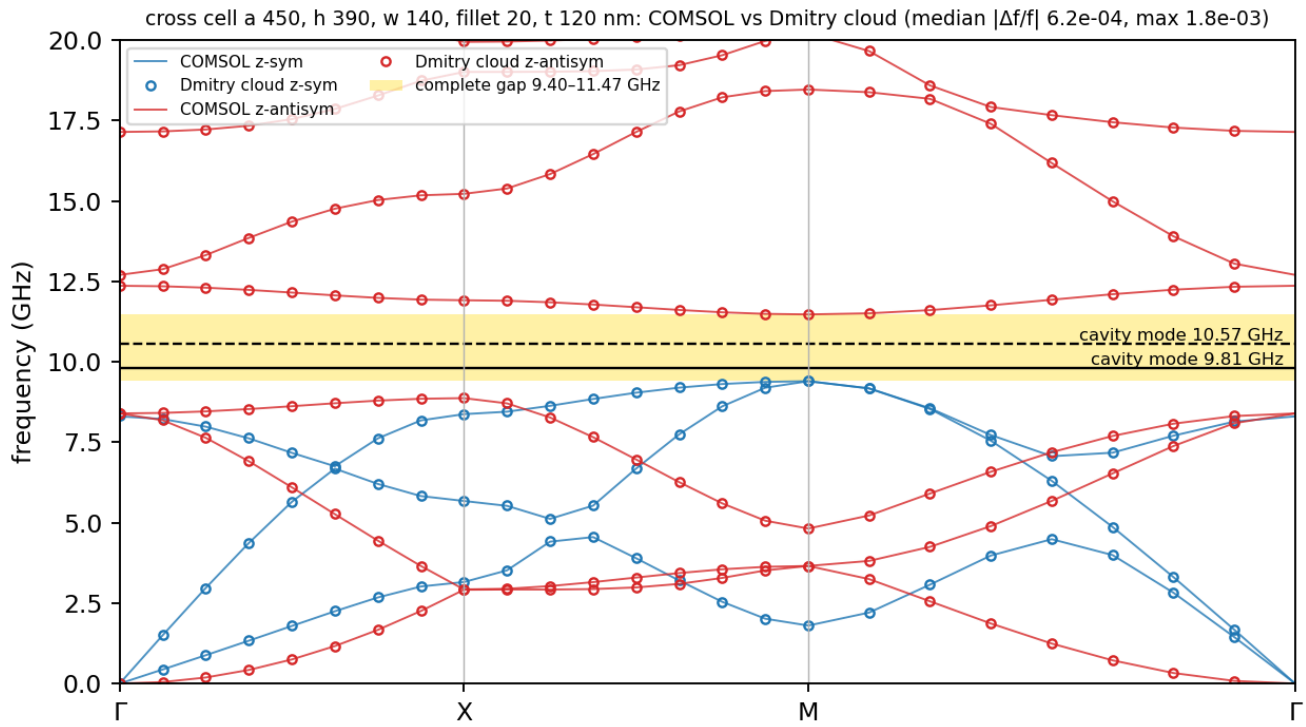
Design on the Dmitry cloud. Square lattice, cross-shaped holes (bar length h, bar width w, fillet 20 nm) in the same 120 nm diamond. 12 cells (sweep-c11edac992d7da32c9a8, Dmitry cloud, P2 elements, z-symmetric + z-antisymmetric, Γ–X–M–Γ, 8 steps per segment): a 350/450/550/650 nm × bridge a − h 60/100 nm at w 100, plus w 70/140 at a 450/550. Cost $0.98 expected ($2.70 ceiling). Every feature ≥ 60 nm. Lattice constant moves the gap (~1/a), bridges set its width, bar width shifts it.

Chosen cell: a 450 nm, h 390 nm, w 140 nm (bridge 60 nm, solid block 310 nm). Complete gap 9.40–11.47 GHz (20 %): it holds both localized modes (9.81 and 10.57 GHz). It is the only cell of the 12 that does.

COMSOL check of that cell (independent model, copy-face Floquet mesh, P2, same k points): gap 9.397–11.472 GHz; 985 modes match the Dmitry cloud to median 6.2e-04, max 1.8e-03 (relative).



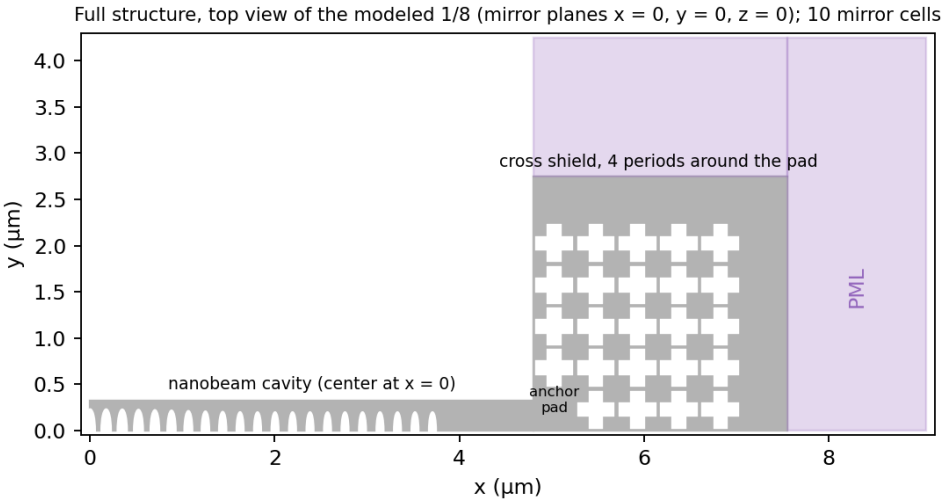
Complete gaps (> 0.8 GHz) of the 12 cross cells (Dmitry cloud). Green: covers both cavity modes.



Bands of the chosen cross cell: COMSOL (lines) vs Dmitry cloud (circles); blue z-symmetric, red z-antisymmetric. Shaded: complete gap. The gap edges are set at M (z-symmetric band below, z-antisymmetric band above).

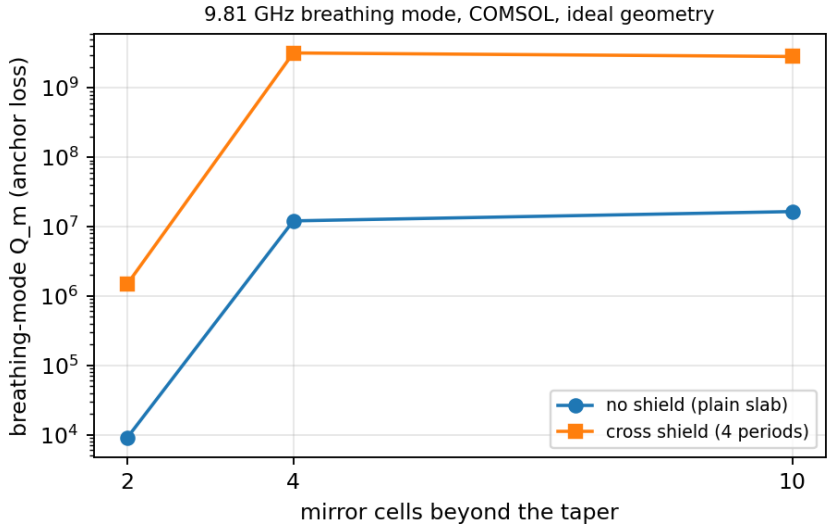
6. Full structure: cavity anchored into the slab, with and without the shield

The beam (best design above, with 10, 4 or 2 mirror cells) runs 1 μm past its last mirror cell and lands on a 120 nm diamond slab; the slab edge next to the beam is free (trench). With the shield, cross holes fill the slab around a 450 × 450 nm solid anchor pad (the pad is enclosed by 4 shield periods in x and y); without it the slab is plain. Outgoing slab waves are absorbed in 1.5 μm PML layers. Mechanics only (the optical mode is microns from the anchor). The breathing-mode Q_m here is anchor (clamping) loss of the ideal symmetric structure.

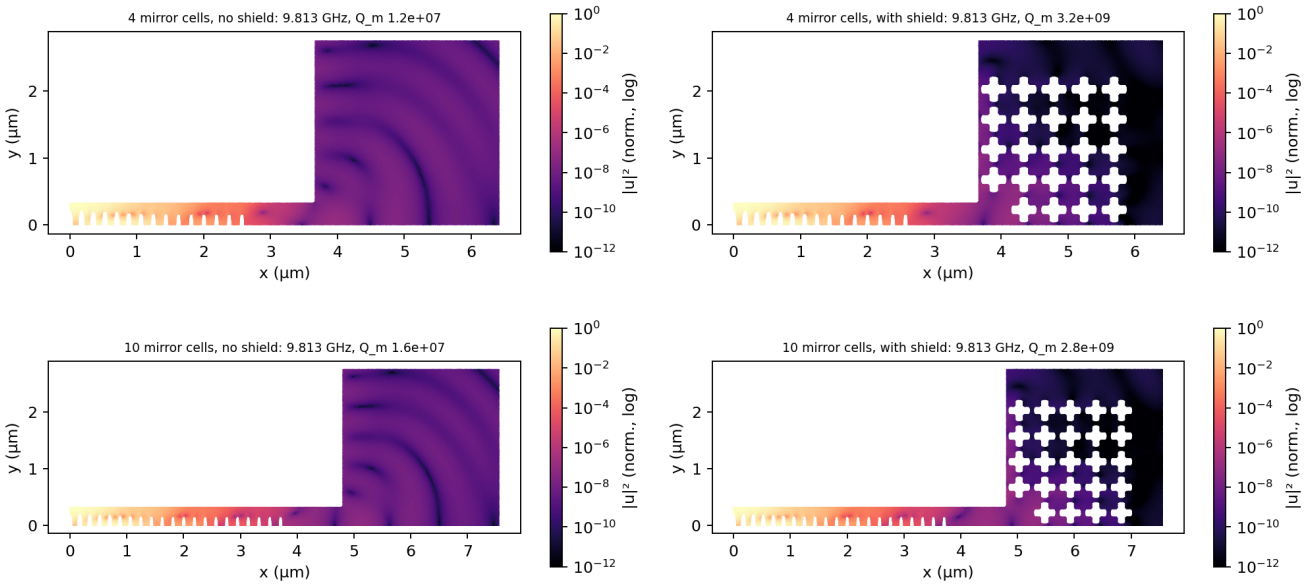


Top view of the modeled eighth (COMSOL). Gray = diamond, white = holes, purple = PML.

mirror cells	no shield: f (GHz)	no shield: Q_m	shield: f (GHz)	shield: Q_m	Q_m ratio
2	9.813	8.96e+03	9.813	1.47e+06	163.7×
4	9.813	1.21e+07	9.813	3.19e+09	263.7×
10	9.813	1.65e+07	9.813	2.83e+09	172.2×



Breathing-mode Q_m vs mirror cells, with and without the shield (COMSOL).



Breathing mode |u|² on the top face (log scale, normalized to the peak), without (left) and with (right) the shield (COMSOL).

Convergence checks

case (10 mirror cells)	f (GHz)	Q_m	vs reference
no shield, reference (PML 1.5 μm, slab mesh 30 nm)	9.8133	1.65e+07	—
no shield, PML 3 μm	9.8130	1.65e+07	1.00×
shield, reference (4 periods, PML 1.5 μm, slab mesh 30 nm)	9.8133	2.83e+09	—
shield, PML 3 μm	9.8130	2.84e+09	1.00×
shield, slab mesh 22 nm	9.8129	2.91e+09	1.03×
shield, 2 periods (same slab size)	9.8133	5.61e+08	0.20×

Findings

The shield raises the breathing-mode Q_m by 160–260× at every mirror length (2, 4, 10 cells). The frequency does not move (9.813 GHz), so the shield does not perturb the cavity mode.

Without the shield, Q_m saturates at ~1.5e7 beyond 4 mirror cells. Adding mirror cells stops helping because the loss is the anchor slab, which radiates waves of every class (rings in the displacement map). Doubling the PML does not change it (1.65e7 both ways), so this is physical anchor loss, not a PML artifact.

With the shield, Q_m is ~3e9 and limited by the shield itself. Doubling the PML: 2.84e9 vs 2.83e9 (0.3 %). Finer slab mesh (22 vs 30 nm): 2.91e9 (+3 %). Two shield periods instead of four: 5.6e8 (5× lower), so the residual loss tunnels through the shield and more periods would raise Q_m further. The mirror length no longer matters beyond 4 cells (3.2e9 at 4, 2.8e9 at 10).

Caveat. Q_m above ~1e9 is an ideal-geometry, anchor-loss-only number. It passes the PML and mesh checks here, but real devices will be limited by disorder (which mixes symmetry classes), material and surface loss, and thermoelastic damping well before 1e9. Measured diamond OMCs reach ~3e4 (Joe 2026) to ~2e6 (Oh 2026). Read the result as: the shield removes anchor loss as the limiting mechanism.

Fabrication note. As modeled, the first shield column leaves a 30 nm strip of diamond at the trench edge (cross center a/2 = 225 nm from the edge, bar half-length 195 nm), below the 60 nm rule. The GDS (best_devices/) reproduces the model exactly. Fix before fabrication, with a re-run: move the lattice at least 30 nm away from the trench, or shorten the edge crosses.

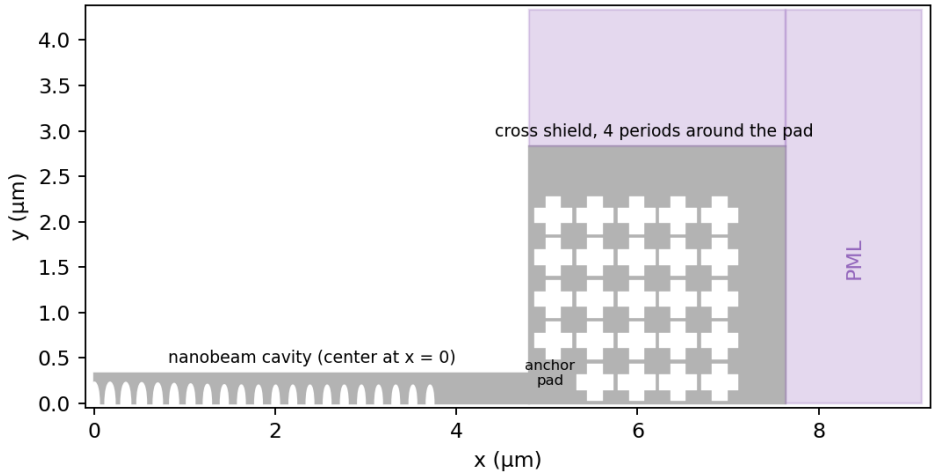
6b. Fabrication-compliant shield: 65 nm minimum feature

The first shield cell has 60 nm bridges, and its first column sits 30 nm from the trench edge. Both are below the 65 nm minimum feature. A second Dmitry-cloud sweep (sweep-5ff274a306306db50948, 6 cells with 65–70 nm bridges, $0.40 expected) found **a 460, h 395, w 150 nm, fillet 20** (bridge 65, bar 150, block 310 nm): complete gap 9.46–11.45 GHz (Dmitry cloud). The lattice is shifted 35 nm away from the trench, so the edge strip is 67.5 nm. The plain-slab comparison uses the same, slightly larger slab (2835 × 2800 nm before the PML).

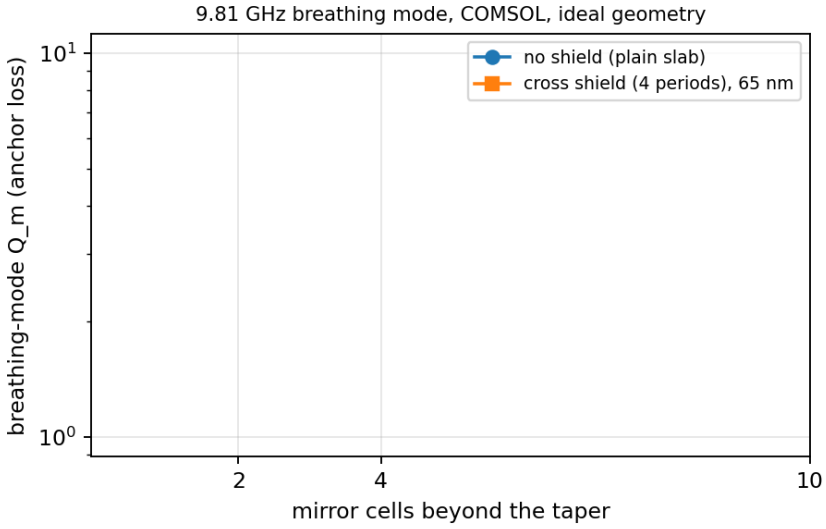
Full structure with the 65 nm shield (COMSOL)

mirror cells	no shield: f (GHz)	no shield: Q_m	shield: f (GHz)	shield: Q_m	Q_m ratio
2	—	—	—	—	—
4	—	—	—	—	—
10	—	—	—	—	—

Full structure, top view of the modeled 1/8 (mirror planes x = 0, y = 0, z = 0); 10 mirror cells; 65 nm shield, lattice 35 nm from the trench



Modeled eighth with the 65 nm shield (lattice 35 nm from the trench).



Q_m vs mirror cells, 65 nm shield vs plain slab.

GDS: best_devices/60nm_w650_d0.086_xi1.79_chi-6.97_gapconf/*_shield.gds (fabrication layout, 65 nm checked); runs/shield_619/gds/ref_*.gds (exactly the COMSOL geometries incl. PML layers, for solver cross-checks).

Not covered here

Fabrication disorder (it breaks the mirror symmetry and is where the shield matters most), sidewall angle, anisotropic (cubic) diamond, material/TLS/thermoelastic loss, optical Q convergence, g0. Full cavity + shield on the Dmitry cloud: its new structure route (GDS input, laghai/nanophotonics-cloud #176) reproduces the plain-anchor COMSOL Q_m; the shielded case is pending on their side.